



OPERATIONAL AMPLIFIER NOISE

Techniques and Tips for Analyzing and Reducing Noise

- Analyze and design low-noise circuits with the help of tricks, rules of thumb, and working examples
- Recognize and troubleshoot specific problems by example, with detailed illustrations and explanations
- Measure, calculate, and predict intrinsic noise levels for quicker design and more effective development

Art Kay

Introduction and Review of Statistics

Noise analysis can be done in the time domain, in the frequency domain, or by using statistical analysis. This chapter introduces these three analysis methods. These methods will be utilized throughout this book.

1.1 Time Domain View of Noise

Noise is most commonly viewed in the time domain. A typical plot of time domain noise is shown in Figure 1.1. In the time domain, noise voltage is on the y -axis and time is on the x -axis. Noise can be viewed in the time domain using an oscilloscope. Figure 1.1 also shows that if you look at this random signal statistically, it can be represented as a Gaussian distribution. The distribution is drawn sideways to help show its relationship with the time domain signal. In Section 1.2, the statistical view is discussed in detail.

Figure 1.1 shows thermal noise in the time domain. Thermal noise is generated by the random motion of electrons in a conductor. Because this motion increases with temperature, the magnitude of thermal noise increases with temperature. Thermal noise can be viewed as a random variation in the voltage present across a component (e.g., resistor). Figure 1.2 gives the equation for finding the root mean square (RMS) thermal noise given resistance temperature and bandwidth.

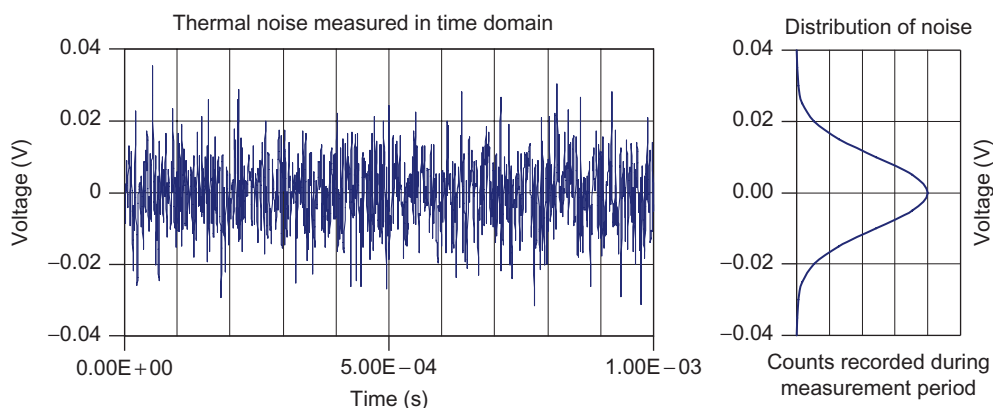


Figure 1.1: Thermal noise shown in the time domain and statistically

$$e_n = \sqrt{4 \cdot k \cdot T \cdot R \cdot \Delta f} \quad (1.1) \text{ Thermal noise equation}$$

where

e_n is the RMS noise voltage

T is temperature in Kelvin (K)

R is resistance in Ohms (Ω)

Δf is noise bandwidth in Hertz (Hz)

k is Boltzmann's constant (J/K)

Note: To convert degrees Celsius to Kelvin

$$T_K = 273.15^\circ\text{C} + T_C$$

Figure 1.2: Thermal noise voltage equation

$$f(x) = \frac{1}{\sigma \cdot \sqrt{2\pi}} \cdot e^{[-(x-\mu)^2/2\sigma^2]} \quad (1.2) \text{ Probability density function for normal (Gaussian) distribution}$$

where

$f(x)$ is the probability that x will be measured at any instant in time

x is the random variable

μ is the mean value

σ is the standard deviation

Figure 1.3: Probability density function for a Gaussian distribution

The important thing to know about the thermal noise equation is that it allows you to find an RMS noise voltage. In many cases, engineers want to know the peak-to-peak noise voltage. In Section 1.2, we will learn some statistical methods that can be used to estimate peak-to-peak noise voltage given the RMS noise value. Section 1.2 also covers other basic statistical methods that are used in noise analysis.

1.2 Statistical View of Noise

Most forms of intrinsic noise have a Gaussian distribution and can be analyzed using statistical methods. For example, statistical methods must be used to calculate the sum of two noise signals or estimate peak-to-peak amplitude. This section gives a short review of some basic statistical methods required to carry out noise analysis.

1.2.1 Probability Density Function

The mathematical equation that forms the normal distribution function is called the “probability density function” (see Figure 1.3). Plotting a histogram of noise voltage measured over a time interval will approximate the shape of this function. Figure 1.4 illustrates a measured noise histogram with the probability distribution function superimposed on it.

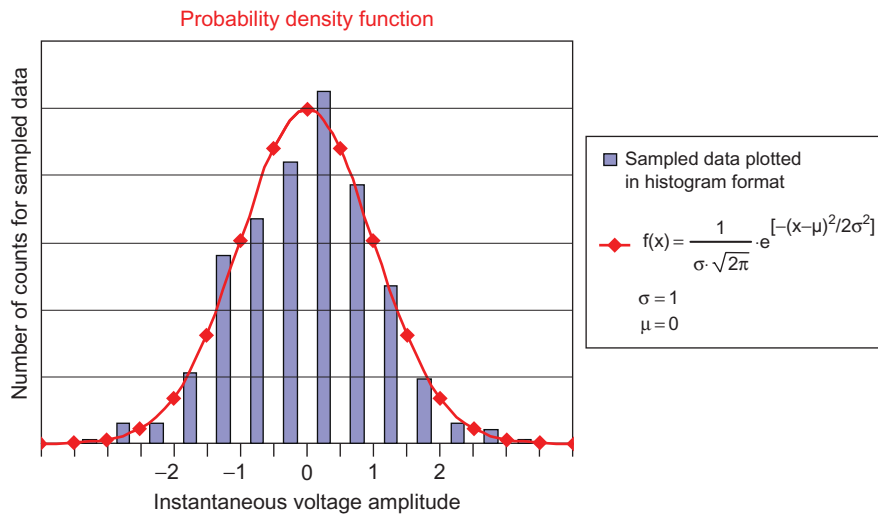


Figure 1.4: Measured distribution with superimposed probability density function

$$P(a < x < b) = \int_a^b f(x) dx = \int_a^b \frac{1}{\sigma \cdot \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \quad (1.3) \text{ Probability distribution function for normal (Gaussian) distribution}$$

where

$P(a < x < b)$ is the probability that x will be in the interval (a, b) in any instant in time. For example, $P(-1 < x < +1) = 0.3$ means that there is a 30% chance that x will be between -1 and 1 for any measurement

x is the random variable. In this case noise voltage

μ is the mean value

σ is the standard deviation

Figure 1.5: Probability distribution function

1.2.2 Probability Distribution Function

The probability distribution function is the integral of the probability density function. This function is very useful because it tells us about the probability of an event that will occur in a given interval (see Figures 1.5 and 1.6). For example, assume that Figure 1.6 is a noise probability distribution function. The function tells us that there is a 30% chance that you will measure a noise voltage between -1 V and $+1$ V [i.e., the interval $(-1, 1)$] at any instant in time.

This probability distribution function is instrumental in helping us translate RMS to peak-to-peak voltage or current noise. Note that the tails of the Gaussian distribution are infinite. This implies that any noise voltage is possible. While this is theoretically true, in practical

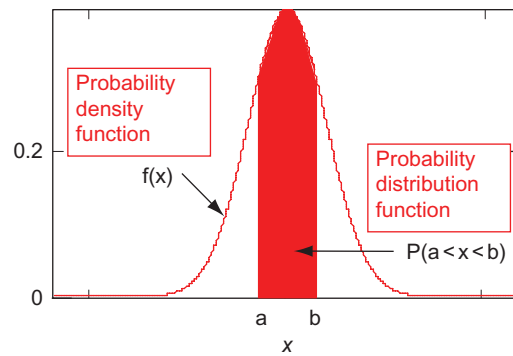


Figure 1.6: Probability density function and probability distribution function

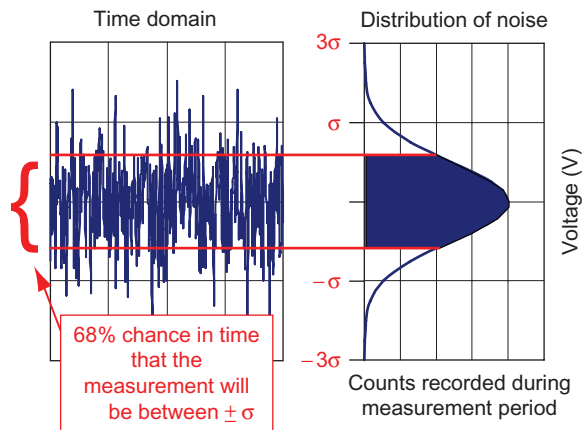


Figure 1.7: Illustrates how standard deviation relates to peak-to-peak

terms the probability that extremely large instantaneous noise voltages are generated is very small. For example, the probability that we measure a noise voltage between -3σ and $+3\sigma$ is 99.7%. In other words, there is only a 0.3% chance of measuring a voltage outside of this interval. For this reason, $\pm 3\sigma$ (i.e., 6σ) is often used to estimate the peak-to-peak value for a noise signal. Note that some engineers use 6.6σ to estimate the peak-to-peak value of noise. There is no agreed-upon standard for this estimation. Figure 1.7 graphically shows how 2σ will catch 68% of the noise. Table 1.1 summarizes the relationship between standard deviation and probability of measuring a noise voltage.

Thus, we have a relationship that allows us to estimate peak-to-peak noise given the standard deviation. In general, however, we want to convert RMS to peak-to-peak amplitude. Often, people assume that the RMS and standard deviation are the same. This is not always the case. The two values are equal only when there is no DC component (the DC component is

Table 1.1: Number of Standard Deviations and Chance of Measuring Voltage

Number of Standard Deviations	Chance of Measuring Voltage (%)
2σ (same as $\pm\sigma$)	68.3
3σ (same as $\pm 1.5\sigma$)	86.6
4σ (same as $\pm 2\sigma$)	95.4
5σ (same as $\pm 2.5\sigma$)	98.8
6σ (same as $\pm 3\sigma$)	99.7
6.6σ (same as $\pm 3.3\sigma$)	99.9

the average value μ). In the case of thermal noise, there is no DC component, so the standard deviation and RMS values are equal.

One way of computing the RMS noise voltage is to measure a large number of discrete points and use statistics to estimate the standard deviation. For example, if you have a large number of samples from an analog-to-digital (A/D) converter, you could use Eqs. (1.4)–(1.6) to compute the mean, standard deviation and RMS of the noise signal. Many software packages such as Microsoft Excel can be used to compute these functions. In fact, Excel has built-in functions for computing standard deviation and mean value. Some test equipment will include these and other built-in mathematical functions. For example, many oscilloscopes include RMS, mean and standard deviation functions.

In general, it is best to use the standard deviation function as apposed to RMS when doing noise computations. Intrinsic noise should not have a DC component. In some practical cases, the instrument measuring the noise may have a DC component. The DC component should not be included in the noise computation because it is not really part of the noise signal. When using the RMS formula on a noise signal with a DC component, the results will be affected by the DC component. The standard deviation formula, however, will eliminate the effects of the DC component.

One final statistical concept to cover is the addition of noise signals. To add two noise signals, we must know if the signals are correlated or uncorrelated. Noise signals from two independent sources are uncorrelated. For example, the noise from two independent resistors or two operational amplifiers (op-amps) is uncorrelated. A noise source can become correlated through a feedback mechanism. Noise-canceling headphones are a good example of the addition of correlated noise sources. They cancel acoustic noise by summing inversely correlated noise. Eq. (1.4) shows how to add correlated noise signals. Note that in the case of the noise-canceling headphones, the correlation factor would be $C = -1$. (see [Figure 1.8](#)).

In most cases, we will add uncorrelated noise sources (see Eq. (1.5)). Adding noise in this form will effectively sum two vectors using the Pythagorean theorem. [Figure 1.9](#) shows the addition graphically. A useful approximation is that if one of these sources is one-third the amplitude of the other, the smaller source can be ignored.

$$e_{nT} = \sqrt{e_{n1}^2 + e_{n2}^2 + 2C \cdot e_{n1} \cdot e_{n2}} \quad (1.4) \text{ Addition of two correlated noise sources}$$

$$e_{nT} = \sqrt{e_{n1}^2 + e_{n2}^2} \quad (1.5) \text{ Addition of two uncorrelated noise sources}$$

where

C is a correlation factor $-1 < C < +1$

e_{n1} is a noise source

e_{n2} is a noise source

Figure 1.8: Addition of random uncorrelated signals

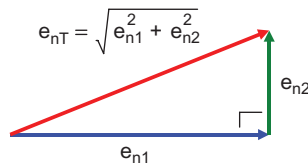


Figure 1.9: Pythagorean theorem for noise

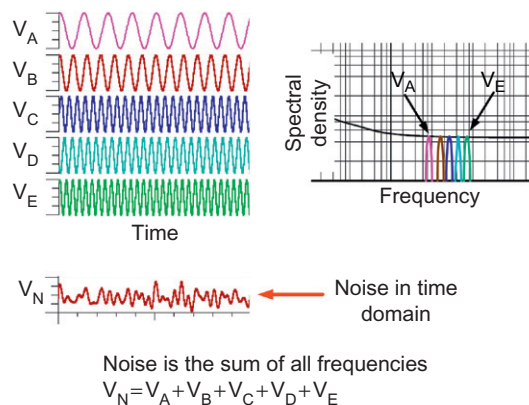


Figure 1.10: Understanding the spectral density curve

1.3 Frequency Domain View of Noise

An important characteristic of noise is its spectral density. Voltage noise spectral density is a measurement of RMS noise voltage per square root hertz (or commonly $nV/\sqrt{\text{Hz}}$). Power spectral density is given in W/Hz . A random noise signal can be thought of as an infinite summation of sine waves at different frequencies. The right side of [Figure 1.10](#) shows how several signals at different frequency add to form a random noise signal. The left side of [Figure 1.10](#) shows the same signals in the frequency domain. Note that each sine wave

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad (1.6) \text{ Mean value}$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \quad (1.7) \text{ Standard deviation}$$

$$\text{RMS} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} \quad (1.8) \text{ RMS}$$

Figure 1.11: Statistical equations for a discrete population

$$e_n = \sqrt{4k \cdot T \cdot R \cdot \Delta f}$$

$$\frac{e_n}{\sqrt{\Delta f}} = \sqrt{4k \cdot T \cdot R} \quad (1.9) \text{ Thermal noise equation}$$

Figure 1.12: Thermal noise equation rearranged into spectral density format

creates an impulse or “spike” in the frequency domain. The example shows just five sine waves combining to form a “random” signal. In reality, noise signals have infinite frequency components. You can imagine an infinite number of impulses in the frequency domain combine to form the spectral density curve.

Earlier, we have learned that the thermal noise of a resistor can be computed using Eq. (1.1). This equation can be rearranged into a spectral density form (see [Figure 1.11](#)). One important characteristic of this noise is that it has a flat spectral density plot (i.e., it has uniform energy at all frequencies). For this reason, thermal noise is sometimes called broadband noise or white noise ([Figure 1.12](#)). The word “white” is used to describe noise with a uniform energy at all frequencies because white light is generated by mixing all colors (wavelengths) with uniform energy. Op-amps also have broadband noise associated with them. Broadband noise is defined as noise that has a flat spectral density plot. [Figure 1.13](#) shows the noise spectral density of a resistor graphed vs. resistance. This plot can be used as a quick way to determine the spectral density of a resistance. Also note that temperature has a very small effect on overall noise.

[Figure 1.14](#) shows two common regions in spectral density curves. In the broadband region the noise spectral density is flat, so the contribution of all the different frequency components is equal. Op-amps also may have a low-frequency noise region that does not have a flat spectral density plot. This noise is called 1/f noise, flicker noise, or low-frequency noise. Typically, the power spectrum of 1/f noise falls at a rate of 1/f. This means that the voltage

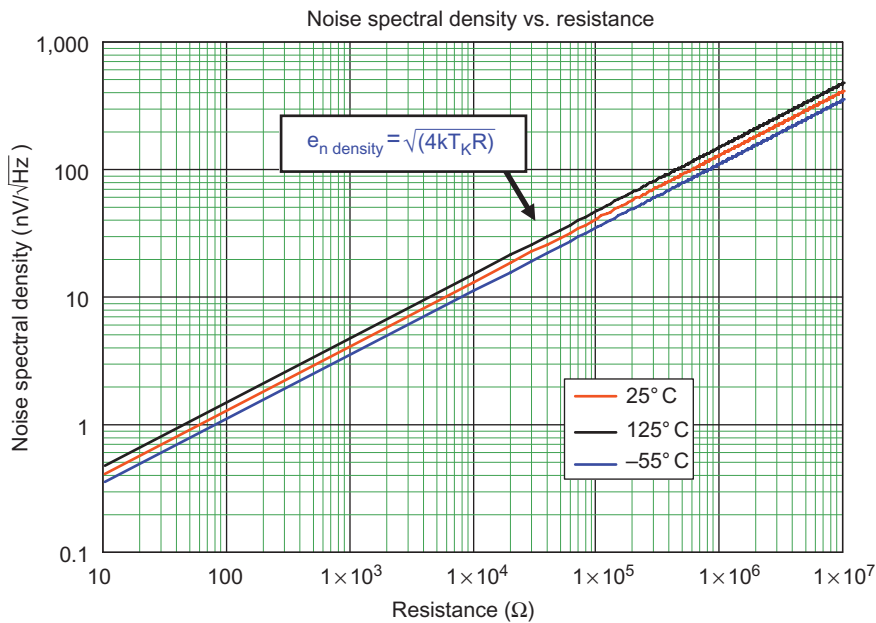


Figure 1.13: Thermal noise equation rearranged into spectral density format

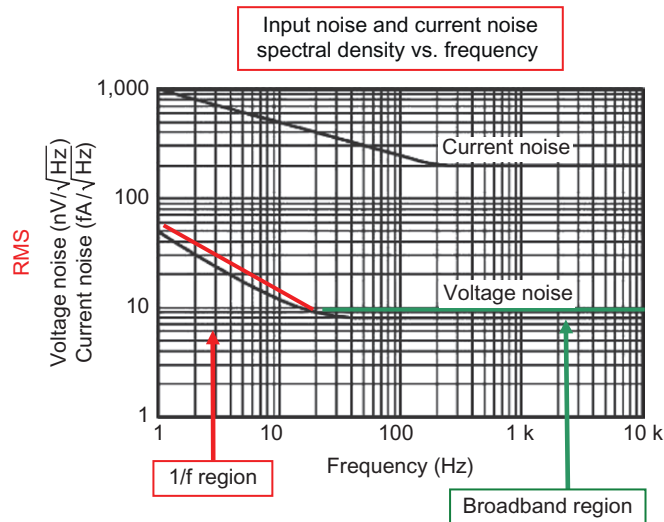


Figure 1.14: Op-amp noise spectral density

spectrum falls at a rate of $1/f^{(1/2)}$. In practice, however, the exponent of the $1/f$ function may deviate slightly. Figure 1.14 shows a typical op-amp spectrum with both a $1/f$ region and a broadband region. Note that the spectral density plot also shows current noise (given in $\text{fA}/\sqrt{\text{Hz}}$).

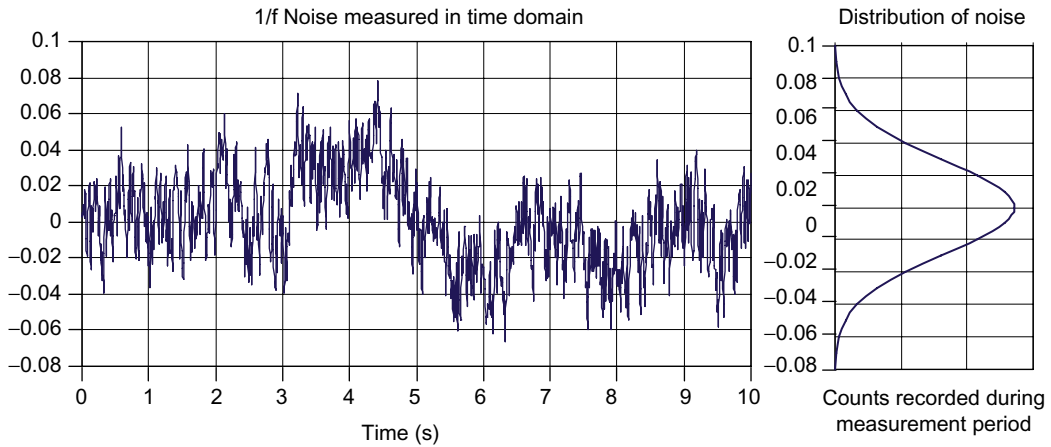


Figure 1.15: 1/f Noise shown in the time domain and statistically

$$\text{Noise power} = \int_{f_1}^{f_2} e_n^2 df \quad (1.10) \text{ RMS noise power over a frequency range. } e_n \text{ is the noise voltage spectral density.}$$

$$\text{Noise voltage} = \sqrt{\int_{f_1}^{f_2} e_n^2 df} \quad (1.11) \text{ RMS noise voltage over a frequency range. } e_n \text{ is the noise voltage spectral density.}$$

Figure 1.16: Converting spectral density to RMS noise

Note that 1/f noise also has a normal distribution, and consequently the mathematics described earlier still applies. Figure 1.15 shows the time domain description of 1/f noise. Note that the x-axis of this graph is given in seconds; this slow change with time is typical for 1/f noise.

1.4 Converting Spectral Density to RMS Noise

A very common noise calculation is to convert spectral density to RMS noise. This is used extensively in op-amp noise calculations. There are three different types of spectral densities to consider: noise power spectral density (unit W/Hz), noise voltage spectral density (unit V/ $\sqrt{\text{Hz}}$), and noise current spectral density (unit A/ $\sqrt{\text{Hz}}$). Noise power spectral density can be converted to RMS power by integrating the spectral density. Power spectral density is defined as voltage or current spectral density squared. Thus, to convert voltage or current noise spectral density to RMS noise, you convert to power (v_n^2 or i_n^2), integrate, and convert back to voltage or current (square root). See Figure 1.16 for details.

A common error that people make when converting voltage spectral density to RMS noise voltage is integrating the voltage spectral density instead of the power spectral density.

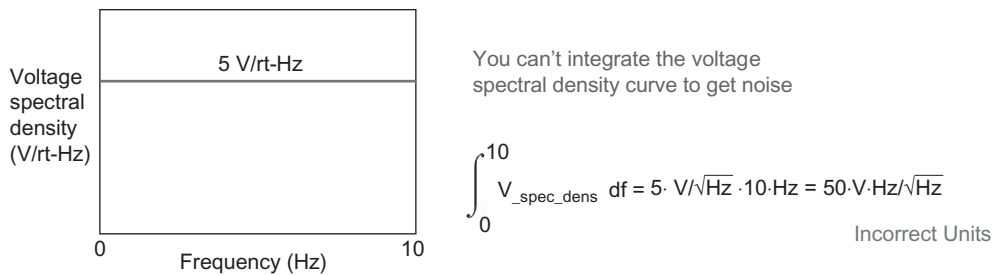


Figure 1.17: Incorrect way to compute noise

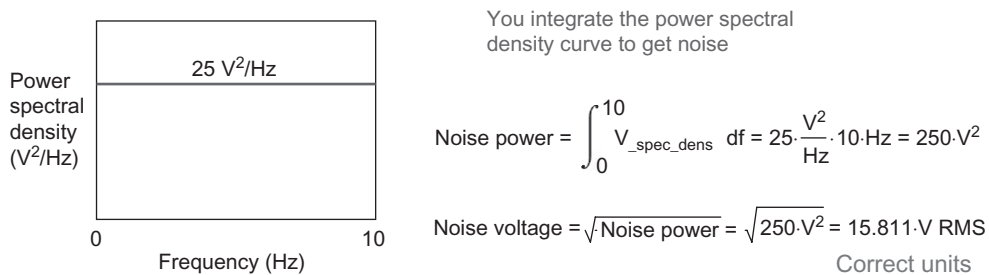


Figure 1.18: Correct way to compute noise

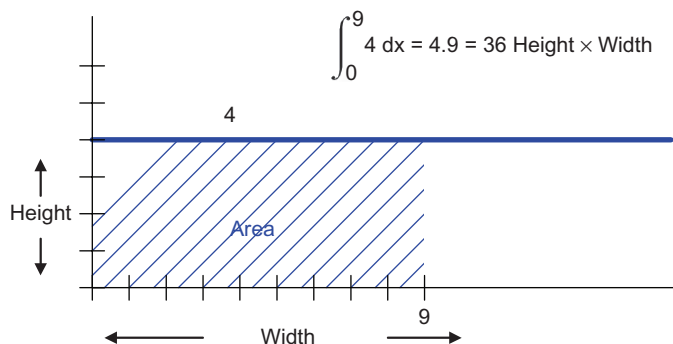


Figure 1.19: Integration computes area under a curve

In Figures 1.17 and 1.18, we will do a dimensional analysis to demonstrate why this does not work. Before doing this, however, we will review the integral. As a quick reminder, the integral function will give the area under a curve. Figure 1.19 shows how a constant function can be integrated by simply multiplying the height by the width (i.e., the area of a rectangle). Considering the integral to be the area of a rectangle simplifies the conversion of spectral density curves to RMS noise values.

Figure 1.17 shows the strange units that result when you attempt to integrate the voltage spectral density curve. Figure 1.18 shows how you can integrate the power spectral density and

convert back to voltage by taking the square root of the result. Note that we get the proper units. Also note that power spectral density is simply the voltage or current spectral density squared (remember $P = V^2/R$ and $P = I^2R$). Thus, by looking at [Figures 1.17 and 1.18](#), you can see why power spectral density must be integrated rather than voltage or current spectral density.

Chapter Summary

- Oscilloscope measurements show noise in the time domain.
 - x -axis is time and y -axis is voltage or current.
- A Gaussian distribution is a statistical view of noise.
 - Standard deviation and RMS are the same if noise does not have a DC component (average value is zero).
 - Six times the standard deviation is a good estimate of peak-to-peak noise.
 - There is a 99.7% probability that noise will be less than six times the standard deviation.
 - Most noise is uncorrelated.
 - Uncorrelated noise is added with the root sum of the square of each noise component.
- The noise spectral density is a frequency domain view of noise.
 - Spectral density has units of $V/\sqrt{\text{Hz}}$ or $A/\sqrt{\text{Hz}}$.
 - White noise is composed of an infinite number of different frequency components with equal energy.
 - Two key regions in a noise spectral density curve are the $1/f$ region and the broadband region.
 - Noise spectral density can be converted to RMS noise by taking the square root of the integral of the noise signal squared.

Questions

- 1.1 Calculate the RMS thermal noise for a $10\text{-k}\Omega$ resistor with a 10-kHz noise bandwidth.
- 1.2 Estimate the peak-to-peak noise for a $10\text{-k}\Omega$ resistor with a 10-kHz noise bandwidth.
- 1.3 Calculate the spectral density for a $10\text{-k}\Omega$ resistor.
- 1.4 A 10mV-RMS and a 5mV-RMS noise source are in series with each other. Assuming the noise sources are random, what is the total noise?
- 1.5 A 10-mV RMS and a 5-mV RMS noise source are in series with each other. Assuming the noise sources are correlated and the correlation factor is -1 , what is the total noise?
- 1.6 A 25-mVpp noise signal is measured. What is the approximate RMS voltage?

Further Reading

Hogg, R.V., Tanis, E.A., 1988, Probability and Statistical Inference, third ed. Macmillan, London.

Motchenbacher, C.D., Connelly, J.A., 1993, Low-Noise Electronic System Design, Wiley-Interscience, New York.

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Introduction to Op-Amp Noise

2.1 Op-Amp Noise Analysis Technique

The goal of op-amp noise analysis technique is to calculate the peak-to-peak output noise of an op-amp circuit based on op-amp data sheet information. As the technique is explained, we will use formulas that apply to most simple op-amp circuits. For more complex circuits, the formulas can help to get a rough idea of the expected noise output. It is possible to develop more accurate formulas for these complex circuits; however, the math would be overly complex. For the complex circuits, it is probably best to use a three-step approach. First, get a rough estimate using the formulas; second, get a more accurate estimate using Spice; and finally, verify your results through measurements.

As the example circuit, we will use a simple noninverting amplifier with a Texas Instruments OPA277 (see Figure 2.1). Our goal is to determine the peak-to-peak output noise. To do this, we have to consider the op-amp current noise, the op-amp voltage noise, and the resistor thermal noise. We will determine the value of these noise sources using the spectral density curves in the data sheet. Also, we will have to consider the gain and bandwidth of the circuit.

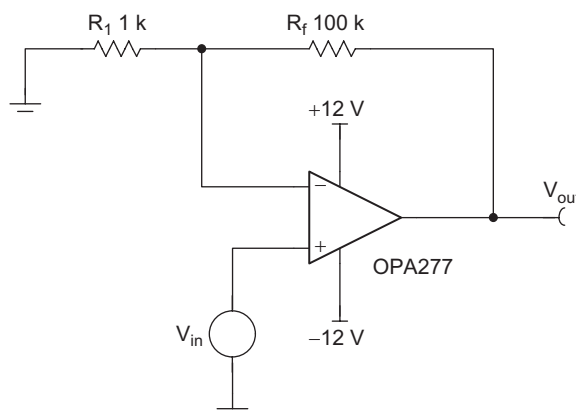


Figure 2.1: Example circuit for noise analysis

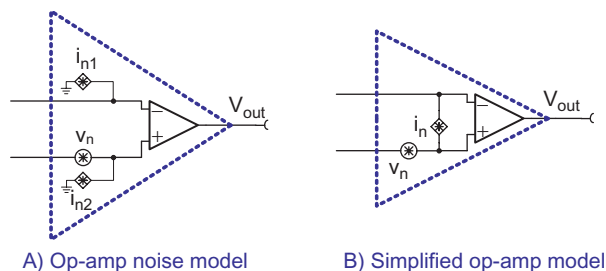


Figure 2.2: Op-amp noise model

2.2 Introducing the Op-Amp Noise Model

Two different models for op-amp noise are shown in Figure 2.2. Figure 2.2A consists of two uncorrelated current noise sources and one voltage noise source connected to the op-amp's inputs. The voltage noise source can be thought of as time-varying input offset voltage component, and the current noise sources can be thought of time-varying bias current components. Note that in some cases, the magnitude of i_{n1} and i_{n2} can be different. In Figure 2.2B, the two current noise sources are combined into a single noise source common to both inputs. The model shown in Figure 2.2B is most commonly used for noise calculations.

2.3 Noise Bandwidth

Integration of the power spectral density curve for the voltage and current spectra will give us the RMS magnitude of the sources in the op-amp model (Figure 2.2). However, the shape of the spectral density curve will contain a $1/f$ region and a broadband region with a low-pass filter (see Figure 2.3). Calculation of the total noise of these two sections will require the use of formulas that were derived using calculus. The results of these two computations are added using root sum of square (RSS) addition for uncorrelated sources, which has been discussed in Chapter 1.

First, we will integrate the broadband region with a low-pass filter. Ideally, the low-pass filter portion of this curve would be a straight vertical line. This is referred to as a brick wall filter. Solving the area under a brick wall filter is easy because it is a rectangle (height $\times$ width). In the real world, we cannot realize a brick wall filter. However, there are a set of constants that can be used to convert real-world filter bandwidth to an equivalent brick wall filter bandwidth for the purpose of the noise calculation. Figure 2.4 compares the theoretical brick wall filter to first-, second-, and third-order filters.

Figure 2.5 shows the equations to convert the real-world filter or the brick wall equivalent. Table 2.1 lists the brick wall conversion factors (K_n) for different filter orders. For example, a

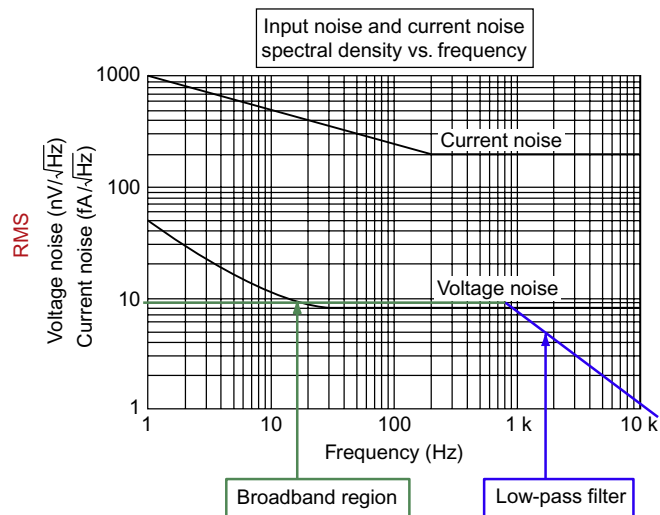


Figure 2.3: Broadband region with filter

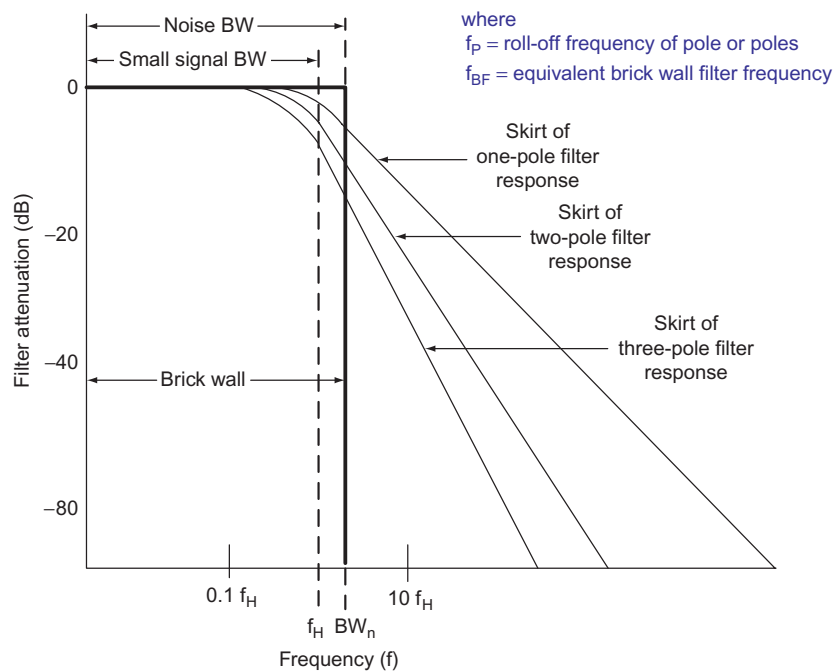


Figure 2.4: Comparison of brick wall filter to real-world filter

$$BW_n = f_H \cdot K_n \quad (2.1) \quad \text{Effective noise bandwidth}$$

where

f_H is the upper-cut frequency

K_n is the brick wall conversion factor

Figure 2.5: Noise bandwidth for simple filter on broadband region

Table 2.1:

Number of Poles in Filter	Kn AC Noise Bandwidth Ratio
1	1.57
2	1.22
3	1.16
4	1.13
5	1.12

$$E_{nBB} = e_{BB} \cdot \sqrt{BW_n} \quad (2.2) \text{ Broadband noise voltage in volts RMS}$$

e_{BB} Broadband noise voltage density (V/ $\sqrt{\text{Hz}}$)

BW_n Noise bandwidth for a given system (Hz)

Figure 2.6: Broadband noise equation

first-order filter bandwidth can be converted to a brick wall filter bandwidth by multiplying by 1.57. The adjusted bandwidth is sometimes referred to as the noise bandwidth. Note that the conversion factor approaches one as the order increases. In other words, higher order filters are a better approximation of a brick wall filter.

Now that we have a formula to convert a real-world filter to its brick wall equivalent, it is a simple matter to integrate the power spectrum. Remember, integrating the power is the voltage spectrum squared. At the end of the integration, the square root is taken to convert back to voltage.

2.4 Broadband RMS Noise Calculation

Eq. (2.2) shown in Figure 2.6 gives the relationship for converting broadband noise spectral density to RMS noise. The broadband noise spectral density (e_{BB}) is taken from the op-amp data sheet. The noise bandwidth is derived using Eq. (2.1) from Table 2.1 (Figures 2.7 and 2.8).

2.5 1/f RMS Noise Calculation

Recall that our goal is to determine the magnitude of the noise source V_n from Figure 2.3. This noise source consists of both broadband noise and 1/f noise. Using Eqs. (2.2) and (2.3), we have been able to compute the broadband component. Now we need to compute the 1/f component. This is done by integrating the power spectrum of the 1/f region of the noise spectral density plot. Figure 2.9 shows this region graphically. The result of the integration is given by Eqs. (2.4) and (2.5). Eq. (2.4) normalizes any noise measurement in the 1/f region to the noise at 1 Hz. In some cases, this number can be read directly from the chart; in other

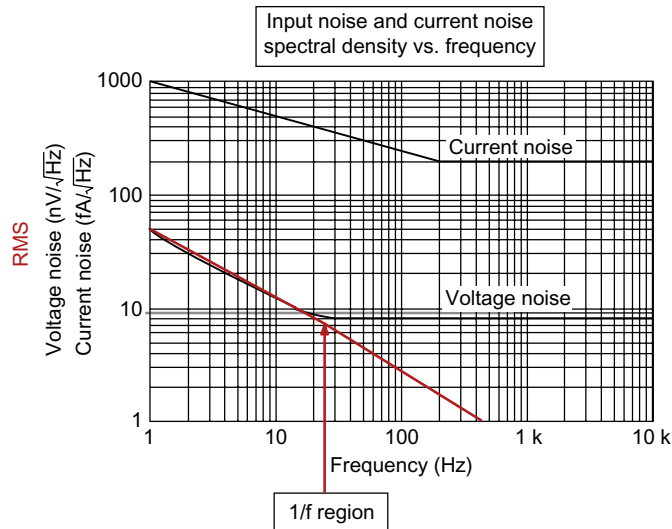


Figure 2.7: 1/f Region

$$e_{\text{f norm}} = e_{\text{at}_f} \cdot \sqrt{f} \quad (2.3) \text{ 1/f noise normalized to 1 Hz}$$

e_{at_f} is voltage noise density selected at a frequency in the 1/f region

f is a frequency in the 1/f region where the noise density is known

Figure 2.8: Noise at 1 Hz (normalized)

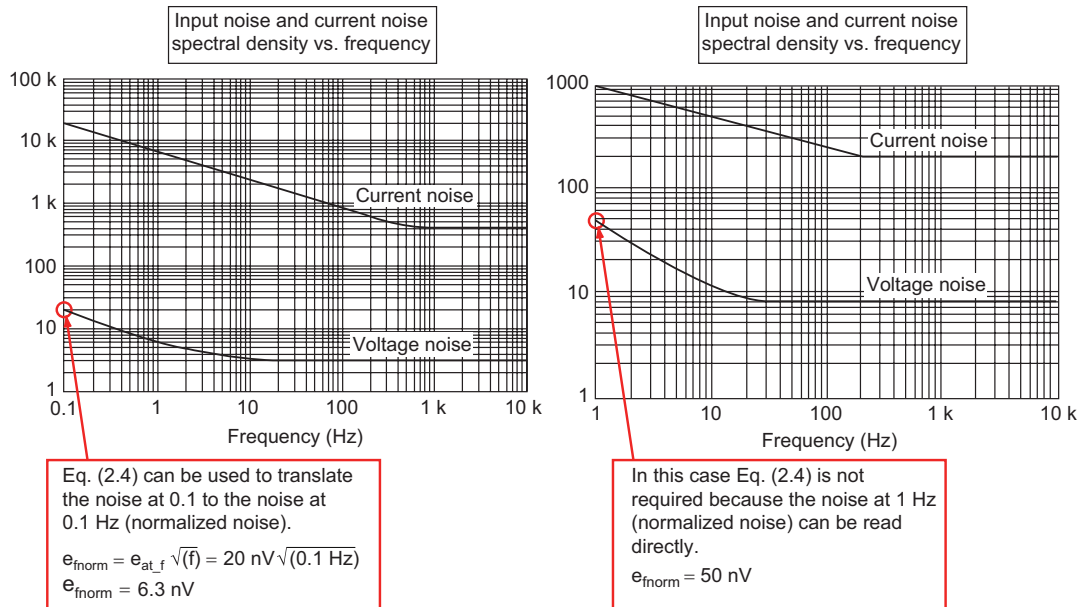


Figure 2.9: Two 1/f normalizing cases

$$E_{nf} = e_{f_{norm}} \cdot \sqrt{\ln \left(\frac{f_H}{f_L} \right)} \quad (2.4) \text{ 1/f noise measured in volts RMS}$$

where

$e_{f_{norm}}$ is normalized noise at 1 Hz from Eq. (2.3)

f_H is upper frequency of operation (use the noise bandwidth BW_n)

f_L is lower frequency of operation (0.1 Hz is typically used)

Figure 2.10: 1/f Noise computation

cases, it is more convenient to use this equation (see Figure 2.10). Eq. (2.5) computes the 1/f noise using the normalized noise, upper noise bandwidth, and lower noise bandwidth. The full derivation is given in Section 2.12.

When considering the 1/f noise you must choose a low-frequency cutoff. This is because the 1/f function is not defined at zero (i.e., 1/0 is undefined). In fact, the noise theoretically goes to infinity when you integrate back to zero hertz. However, you should consider that very low frequencies correspond to long times. For example, 0.1 Hz corresponds to 10 s, and 0.001 Hz corresponds to 1000 s. For extremely low frequencies, the corresponding time could be years (e.g., 10 nHz = 3 years). The greater the frequency interval that is integrated, the larger the resultant noise. Keep in mind, however, that extremely low frequency noise measurements must be made over a long period of time. These phenomena will be discussed in greater detail in Chapter 8. For now, please note that 0.1 Hz is often used for the lower cutoff frequency of the 1/f calculation.

2.6 Combining Flicker and Broadband Noise

Now we have both the broadband and the 1/f noise magnitude. We must add these noise sources using the formula for uncorrelated noise sources given in Chapter 1 (see Eq. (2.5) and Eq. (1.8)).

A common concern that engineers have when considering this analysis technique is that they feel that the 1/f noise and broadband noise should be integrated in two separate regions. In other words, they believe that adding noise in this region will create an error because the 1/f noise will add with the broadband noise outside of the “1/f region.” In reality the 1/f region extends across all frequencies as does the broadband region. Note that the noise spectrum is shown on a log chart, so the 1/f region has little impact after it drops below the broadband curve. The only region where the combination of the two curves is obvious is near where they combine (often called the 1/f corner frequency). In this region, we can see that the two sections combine as is described by our mathematical model. Figure 2.12 illustrates how the two regions actually overlap as well as giving some relative magnitudes.

$$E_{n_v} = \sqrt{E_{nf}^2 + E_{nBB}^2} \quad (2.5) \text{ Total RMS voltage noise}$$

where

E_{nf} is total RMS 1/f noise

E_{nBB} is total RMS broadband noise

Figure 2.11: Addition of 1/f and broadband noise

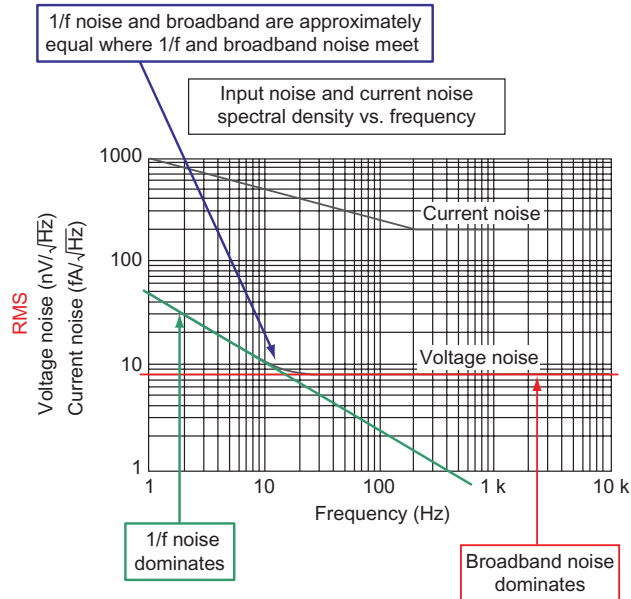


Figure 2.12: 1/f Noise and broadband noise regions overlap

2.7 Noise Model for Example Circuit

In Chapter 1, we have developed a method for converting the noise spectral density curves from a product data sheet to noise sources in an op-amp model. In this chapter, we will learn how to use the model to compute the total output noise for a simple op-amp circuit. The total noise referred-to-the-input (RTI) will contain noise from the op-amp voltage noise source, noise from the op-amp current noise source, and resistor noise. This combined noise source will be multiplied by the op-amp “noise gain.” Figure 2.13 shows all the different sources needing to be combined and multiplied by the noise gain.

2.8 Noise Gain

Noise gain is the gain that the op-amp circuit has to the total noise RTI. In some cases this is not equivalent to the signal gain. Figure 2.14 shows an example where the signal gain is one and the noise gain is two. The V_n source represents contributions of noise from several sources; it is positioned on the noninverting input (“+ve” input) of the op-amp circuit. Note

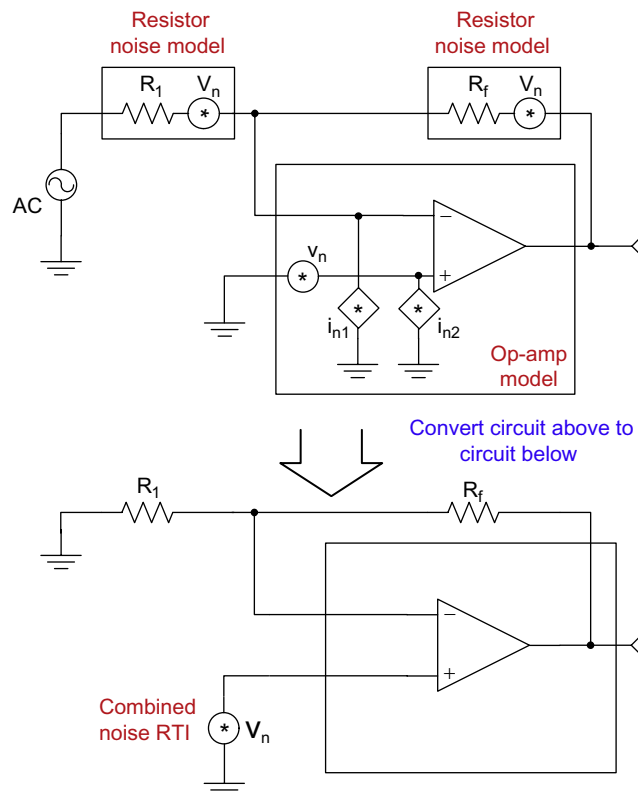


Figure 2.13: Combining the noise sources

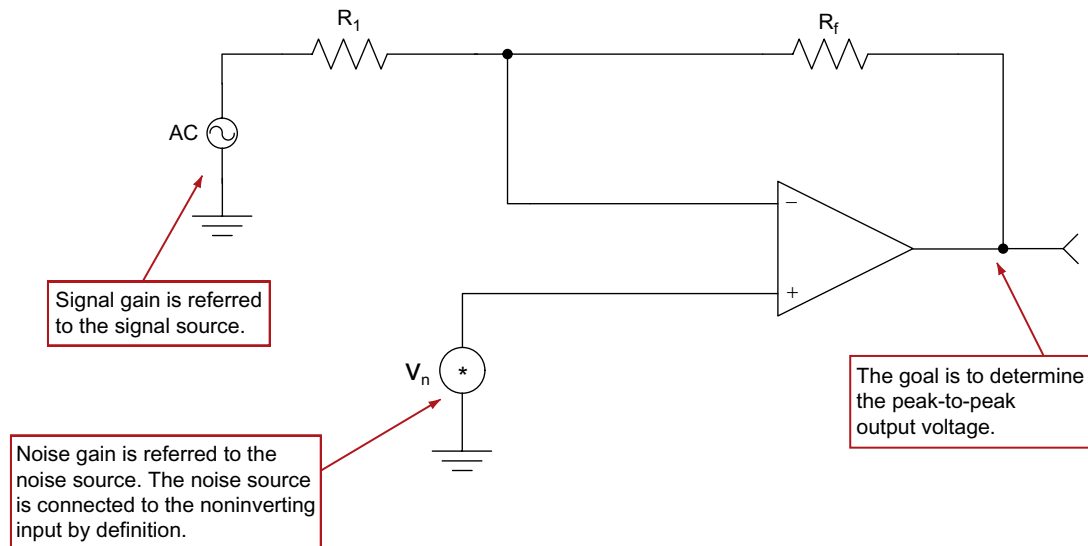


Figure 2.14: Noise gain vs. signal gain

$$\text{Noise gain} = \frac{R_f}{R_1} + 1 \quad (2.6) \text{ Noise gain is the gain seen by the noise voltage source}$$

Figure 2.15: Noise gain for simple op-amp circuit

$$I_{nBB} = i_{nBB} \cdot \sqrt{BW_n} \quad (2.7) \text{ Broadband RMS noise current}$$

$$R_{eq} = \frac{R_f \cdot R_1}{R_f + R_1} \quad (2.8) \text{ Equivalent resistance seen by current noise source}$$

$$i_{f_{norm}} = i_{at_f} \cdot \sqrt{f} \quad (2.9) \text{ 1/f noise normalized to 1 Hz}$$

$$I_{nf} = i_{f_{norm}} \cdot \sqrt{\ln \left(\frac{f_H}{f_L} \right)} \quad (2.10) \text{ 1/f noise measured in amps RMS}$$

$$I_n = \sqrt{I_{nf}^2 + I_{nBB}^2} \quad (2.11) \text{ Total RMS noise from current noise}$$

$$E_{n_i} = I_n \cdot R_{eq} \quad (2.12) \text{ RMS noise voltage from current noise}$$

where

i_{nBB} is current noise broadband spectral density

i_{at_f} is current noise spectral density in 1/f region at f

f is frequency that the current spectral density i_{at_f} is measured at

BW_n is noise bandwidth (2.1)

R_f is feedback resistor

R_1 is input resistor

Figure 2.16: Converting current noise to voltage noise (RTI) for simple op-amp

that it is a common engineering practice to lump all the noise sources to a common source at the noninverting input. Our end goal is to compute noise referred-to-the-output (RTO) of the op-amp circuit (Figure 2.15).

2.9 Converting Current Noise to Voltage Noise

From the previous chapter we know how to compute the voltage noise input, but how do we convert the current noise sources to a voltage noise source? One way of doing this is to carry out an independent nodal analysis for each current source and use superposition to sum the results. Be careful to make sure that the results from each current source is added using RSS. Eqs. (2.7)–(2.12) in Figure 2.16 allow you to convert current noise to an equivalent voltage noise source for a simple op-amp circuit. Figure 2.17 shows the noise equivalent circuit.

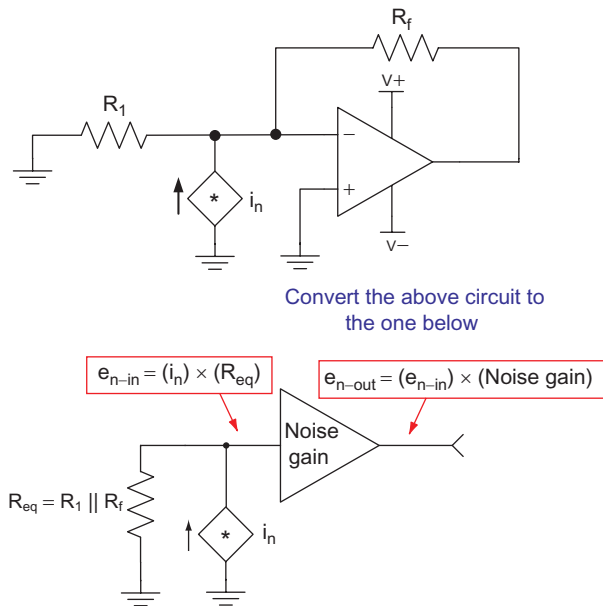


Figure 2.17: Converting current noise to voltage noise (equivalent circuit)

$$E_{n_r} = \sqrt{4kT \cdot R_{eq} \cdot BW_n} \quad (2.13) \text{ Total RMS noise from thermal noise of op-amp feedback network}$$

$$R_{eq} = \frac{R_f \cdot R_1}{R_f + R_1} \quad (2.8) \text{ Equivalent resistance seen by current noise source}$$

where

k is Boltzmann's constant (1.38×10^{-23} J/K)

T is temperature in Kelvin (K)

R_{eq} is equivalent resistance

BW_n is noise bandwidth

Figure 2.18: Thermal noise RTI for simple op-amp circuit

2.10 Including the Effect of Thermal Noise

Another thing that must be considered is the thermal voltage noise from the resistors in the op-amp circuit. Figure 2.18 gives the equations for thermal noise. From an AC perspective, R_f and R_1 are in parallel, so the equivalent resistance given in Eq. (2.8) can be used to find the thermal noise equivalent resistance. Eq. (2.13) uses the equivalent resistance and noise bandwidth to compute thermal noise. This input referred

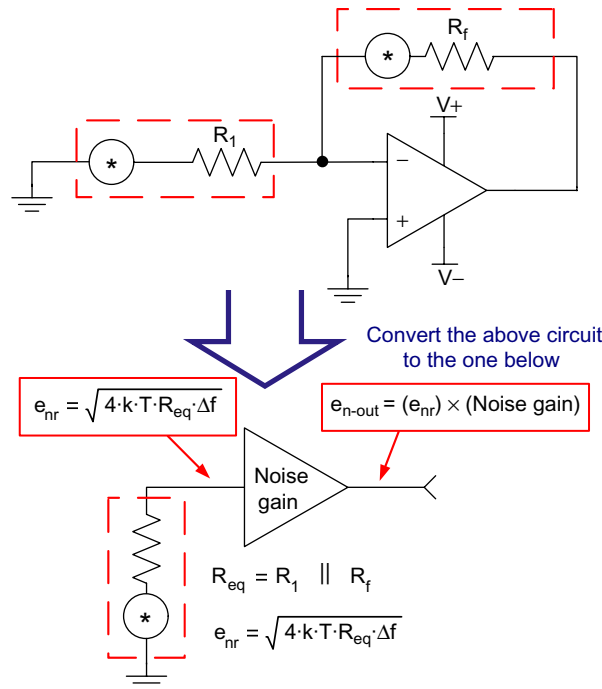


Figure 2.19: Thermal noise RTI for simple op-amp circuit (equivalent circuit)

$$E_{n_in} = \sqrt{E_{n_i}^2 + E_{n_v}^2 + E_{n_r}^2} \quad (2.14) \text{ Total RMS noise referred to the input}$$

$$E_{n_out} = E_{n_in} \cdot \text{Noise_gain} \quad (2.15) \text{ Total RMS noise referred to the output}$$

$$E_{n_out_pp} = E_{n_out} \cdot 6.0 \quad (2.16) \text{ Peak-to-peak output voltage}$$

Figure 2.20: Conversion to peak-to-peak noise

thermal noise source is expressed as an equivalent resistor. Figure 2.19 shows the noise equivalent circuit.

2.11 Combining All the Noise Sources and Computing Peak-to-Peak Output Noise

The final step to compute output noise is to combine all the noise sources and multiply by the noise gain. This RMS noise is used to estimate the peak-to-peak noise by multiplying by 6. Recall from Chapter 1 that there is a 99.7% chance that any instantaneous noise measurement will be less than six times the RMS noise. Eqs. (2.14)–(2.16) from Figure 2.20 summarize this final step.

2.12 Derivation of Key Noise Formulas

Figure 2.21 shows the derivation of the brick wall correction factor used when computing noise bandwidth. This derivation involves integrating the power spectral density and taking the square root of the result.

$$E_{\text{RMS}}^2 = \int_{f_1}^{f_2} e_n^2 \cdot (|G|)^2 df$$

where

E_{RMS} is total RMS noise from f_1 to f_2 in V_{RMS}

e_n is magnitude of noise spectral density at f_1 in $V/\sqrt{\text{Hz}}$

G is gain function for a single pole filter

$$G = \frac{1}{1 + \frac{j\omega}{\omega_p}} \quad |G| = \frac{1}{\sqrt{1^2 + \frac{f^2}{f_p^2}}} \quad (|G|)^2 = \frac{1}{1^2 + \frac{f^2}{f_p^2}}$$

$$E_{\text{RMS}}^2 = \int_{f_1}^{f_2} e_n^2 \cdot \frac{1}{1 + \frac{f^2}{f_p^2}} df = \int_{f_1}^{f_2} e_n^2 \cdot \frac{f_p^2}{f_p^2 + f^2} df$$

$$E_{\text{RMS}}^2 = e_n^2 \cdot f_p \cdot \text{atan}\left(\frac{f_2}{f_p}\right) - e_n^2 \cdot f_p \cdot \text{atan}\left(\frac{f_1}{f_p}\right)$$

Let $f_1 = 0$, $f_2 = \infty$

$$E_{\text{RMS}}^2 = e_n^2 \cdot f_p \cdot \text{atan}(\infty) - e_n^2 \cdot f_p \cdot \text{atan}\left(\frac{f_1}{f_p}\right) = e_n^2 \cdot f_p \cdot \frac{\pi}{2}$$

$$E_{\text{RMS}}^2 = e_n^2 \cdot f_p \cdot \frac{\pi}{2}$$

$$E_{\text{RMS}} = \sqrt{e_n^2 \cdot f_p \cdot \frac{\pi}{2}}$$

Note that $\pi/2$ is $K_n = 1.57$ from Table 2.1

Figure 2.21: Derivation of brick wall correction factor

$$E_n = \frac{e_{\text{normal}}}{\sqrt{f}}$$

$$E_n^2 = \frac{e_{\text{normal}}^2}{(\sqrt{f})^2} = \frac{e_{\text{normal}}^2}{f}$$

$$E_{nf}^2 = \int_{f_L}^{f_H} \frac{e_{\text{normal}}^2}{f} df = e_{\text{normal}}^2 \cdot (\ln(f_H) - \ln(f_L))$$

$$E_{nf}^2 = e_{\text{normal}}^2 \cdot \ln\left(\frac{f_H}{f_L}\right)$$

$$E_{nf} = e_{\text{normal}} \cdot \sqrt{\ln\left(\frac{f_H}{f_L}\right)} \quad (2.4) \text{ Equation for calculating RMS noise in } 1/f \text{ region}$$

Figure 2.22: Derivation of 1/f region

$$\text{Output_noise}^2 = \sqrt{e_{n1}^2 + e_{n2}^2} = \sqrt{\left(e_{n1} \cdot \frac{R_f}{R_1}\right)^2 + e_{n2}^2}$$

$$\text{Let } \beta = 4 \cdot kT \cdot BW$$

$$\text{Output_noise} = \sqrt{\left(\sqrt{\beta \cdot R_f} \cdot \frac{R_f}{R_1}\right)^2 + \left(\sqrt{\beta \cdot R_f}\right)^2}$$

$$\text{Input_noise} = \frac{\sqrt{\left(\sqrt{\beta \cdot R_f} \cdot \frac{R_f}{R_1}\right)^2 + \left(\sqrt{\beta \cdot R_f}\right)^2}}{\frac{R_f}{R_1} + 1}$$

$$\text{Input_noise}^2 = \frac{\beta \cdot \frac{R_f^2}{R_1} + \beta \cdot R_f}{\left(\frac{R_f + R_1}{R_1}\right)^2} = \frac{\beta \cdot R_f^2 \cdot R_1 + \beta \cdot R_f \cdot R_1^2}{(R_f + R_1)^2}$$

$$\text{Input_noise} = \frac{\sqrt{\beta \cdot R_f^2 \cdot R_1 + \beta \cdot R_f \cdot R_1^2}}{(R_f + R_1)^2} = \sqrt{\beta \cdot \frac{R_f \cdot R_1}{R_f + R_1}}$$

$$\text{Input_noise} = \sqrt{4 \cdot kT \cdot BW \cdot \left(\frac{R_f \cdot R_1}{R_f + R_1}\right)} \quad \text{Eq. to noise of } R_f \parallel R_1$$

Figure 2.23: The equivalent noise resistance

Figure 2.22 shows the derivation for the $1/f$ noise region. This derivation involves integrating the power spectral density and taking the square root of the result.

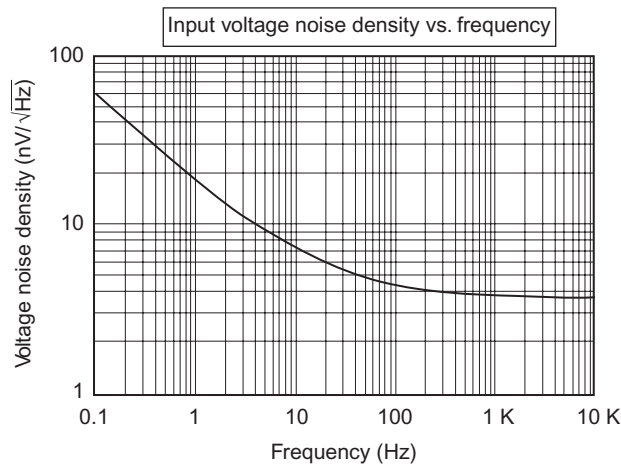
This calculation shows the combination of the two thermal noise sources from each resistor in an op-amp circuit. The two sources are combined using superposition and the final result is algebraically rearranged to prove that the equivalent resistance is the parallel combination of R_f and R_1 . This makes sense because the two resistors are in parallel from an AC perspective and noise is an AC signal (Figure 2.23).

Chapter Summary

- The simplified op-amp noise model contains current noise and voltage noise.
- Noise bandwidth of a simple filter can be calculated by multiplying the signal bandwidth (i.e., -3 -dB bandwidth) by a brick wall correction factor.
- Noise calculations involve looking at the effects of $1/f$ noise and broadband noise separately. The two components are added using the root sum of squares.
- The final noise calculation takes into account current noise, voltage noise, and thermal noise.

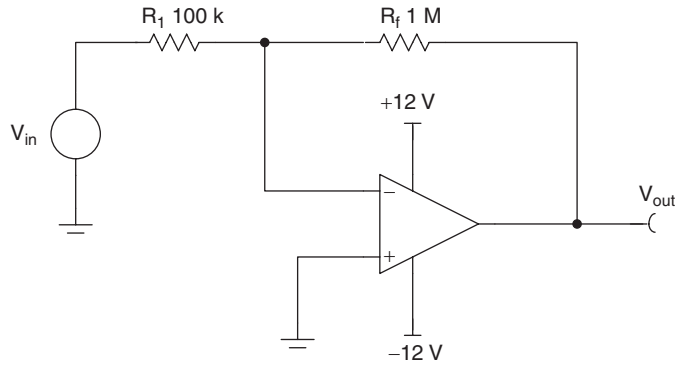
Questions

- 2.1 Given a signal bandwidth of 1 kHz, what is the noise bandwidth? Assume a first-order filter.
- 2.2 Given the spectral density curve below:
 - a. What is the total noise contribution of the $1/f$ region? Assume $f_L = 0.1$ Hz and $f_H = 1$ kHz.
 - b. What is the total noise contribution of the broadband region? Assume the signal bandwidth is 1 kHz and the filter is first order.



2.3 For the circuit below:

- What is the thermal noise? Assume $BW_n = 5 \text{ kHz}$.
- What is the current noise referred-to-the-input. Assume $i_n = 5 \text{ fA}/\sqrt{\text{Hz}}$, and $BW_n = 5 \text{ kHz}$.



- 2.4 Assume the total noise at the output of an op-amp contains $E_{n-i} = 100 \mu\text{V}_{\text{RMS}}$, $E_{n-v} = 150 \mu\text{V}_{\text{RMS}}$, and $E_{n-r} = 75 \mu\text{V}_{\text{RMS}}$. What is the total noise?

Further Reading

Robert V. Hogg, and Elliot A. Tanis. Probability and Statistical Inference, third ed. Macmillan, London.
 C. D. Motchenbacher, and J. A. Connelly, 1993. Low-Noise Electronic System Design. Wiley-Interscience, New York.

Op-Amp Noise Example Calculations

This chapter covers two detailed noise hand calculations. These calculations use the equations developed in Chapter 2. The concept of the gain bandwidth limiting the op-amp frequency response is introduced. Also, a multiple-stage amplifier example shows how the input stage will be the dominant noise source.

3.1 Example Calculation #1: OPA627 Noninverting Amplifier

At this point, we are finally ready to go through a real-world example. Sometimes engineers are overwhelmed by the amount of work required to get to this point. In fact, it is possible to use simulation software to do some of this difficult work for you. However, it is important to have an understanding of the theoretical background because it will give you a more intuitive understanding of how noise works. Furthermore, you should always do a quick “back-of-the-envelope” calculation before you simulate a circuit so that you know if your simulation result is correct. In Chapter 4, we will discuss how to do this analysis using a Spice simulator package.

Figure 3.1 illustrates the simple op-amp configuration that will be used for this analysis example. Note that the specifications used in this example were taken from the OPA627 data sheet. This data sheet can be downloaded from the Texas Instruments website (<http://www.ti.com>).

3.2 Compute the Noise Bandwidth

The first step to this analysis is to determine the noise gain and noise bandwidth for this circuit. The noise gain is given by Eq. (3.2) ($\text{Noise_Gain} = R_f/R_1 + 1 = 100\text{k}/1\text{k} + 1 = 101$). The signal bandwidth is limited by the closed loop bandwidth of the op-amp. Using the gain bandwidth product from the data sheet, the closed loop bandwidth can be determined using Eq. (3.1). If the gain bandwidth product is not listed in the data sheet, use the unity-gain bandwidth specification. The unity-gain bandwidth is the same as gain bandwidth for unity-gain stable amplifiers. Figure 3.2 shows the equations used to calculate bandwidth. Figure 3.5 shows graphically how bandwidth is limited by Aol.

3.3 Get Key Noise Specifications from the Data Sheet

The next part of the analysis is to get the broadband and 1/f noise spectral density specification from the data sheet. The specification is sometimes shown graphically

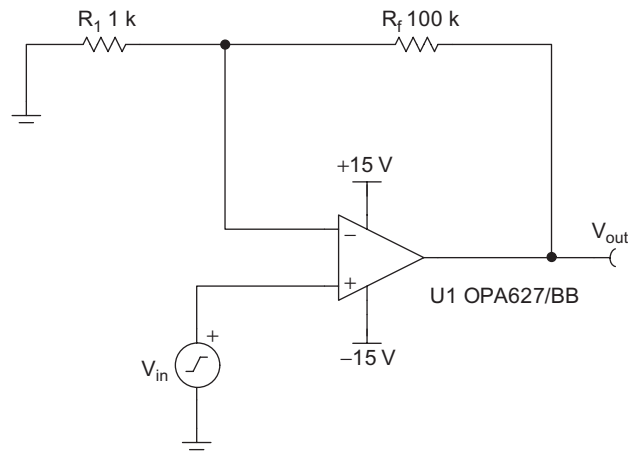


Figure 3.1: Example circuit

$$\text{Closed_Loop_Bandwidth} = \frac{\text{Gain_Bandwidth_Product}}{\text{Noise_Gain}} \quad (3.1) \text{ Op-amp bandwidth limit}$$

$$\text{Closed_Loop_Bandwidth} = \frac{16 \text{ MHz}}{101} = 158 \text{ kHz}$$

Figure 3.2: Closed loop bandwidth for simple noninverting amp

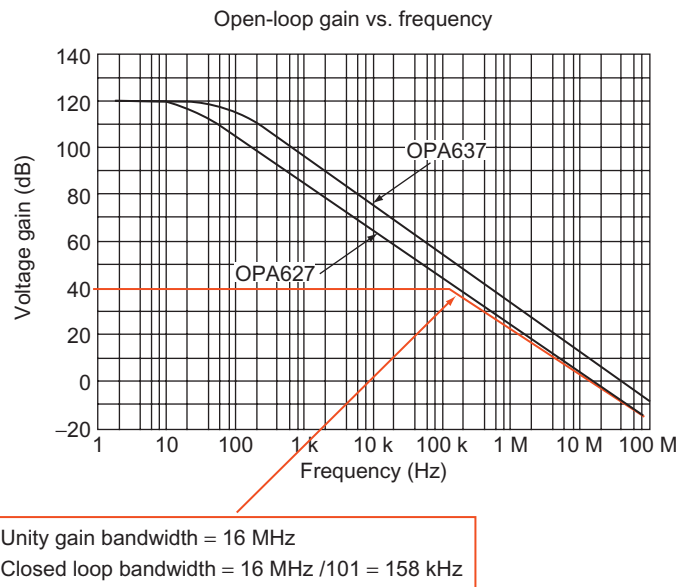


Figure 3.3: Closed loop bandwidth for simple noninverting amp

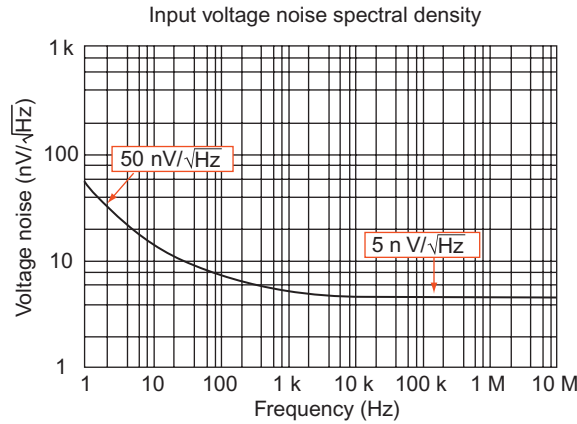


Figure 3.4: OPA627 noise spectral density specification to be used in calculations

Computer total op-amp voltage noise contribution:

Broadband voltage noise component

$$BW_n = f_H \cdot K_n \quad (2.1) \text{ Noise bandwidth}$$

$$BW_n = (158 \text{ kHz}) \cdot (1.57) = 248 \text{ kHz}$$

$$E_{nBB} = e_{BB} \cdot \sqrt{BW_n} \quad (2.2) \text{ Total broadband noise from voltage noise}$$

$$E_{nBB} = \left(5 \frac{\text{nV}}{\sqrt{\text{Hz}}} \right) \cdot \sqrt{248 \text{ kHz}} = 2490 \text{ nV}_{\text{RMS}}$$

1/f Voltage noise component:

$$e_{\text{fnorm}} = e_{\text{at}_f} \cdot \sqrt{f} \quad (2.3) \text{ 1/f Noise normalized to 1 Hz}$$

$$e_{\text{fnorm}} = \left(50 \frac{\text{nV}}{\sqrt{\text{Hz}}} \right) \cdot \sqrt{1 \text{ Hz}} = 50 \text{ nV}$$

$$E_{\text{nf}} = e_{\text{fnorm}} \cdot \sqrt{\ln \left(\frac{f_H}{f_L} \right)} \quad (2.4) \text{ Total 1/f noise from voltage noise}$$

$$E_{\text{nf}} = (50 \text{ nV}) \cdot \sqrt{\ln \left(\frac{248 \text{ kHz}}{0.1 \text{ Hz}} \right)} = 191.8 \text{ nV}_{\text{RMS}}$$

Total voltage noise (referred to the input of the amplifier):

$$E_{n_V} = \sqrt{E_{\text{nf}}^2 + E_{nBB}^2} \quad (2.5) \text{ Total RMS noise from voltage noise}$$

$$E_{n_V} = \sqrt{(2490 \text{ nV}_{\text{RMS}})^2 + (191.8 \text{ nV}_{\text{RMS}})^2} = 2497 \text{ nV}_{\text{RMS}}$$

Figure 3.5: Computation of the magnitude of the voltage noise referred-to-the-input

Table 3.1: OPA627 Noise Spectral Density Specification

	OPA627AM, AP, AU OPA637AM, AP, AU			Units
	Min	Typ	Max	
Input voltage noise				
Noise density: $f = 10 \text{ Hz}$		20		$\text{nV}/\sqrt{\text{Hz}}$
$f = 100 \text{ Hz}$		10		$\text{nV}/\sqrt{\text{Hz}}$
$f = 1 \text{ kHz}$		5.6		$\text{nV}/\sqrt{\text{Hz}}$
$f = 10 \text{ kHz}$		4.8		$\text{nV}/\sqrt{\text{Hz}}$
Voltage noise, $\text{BW} = 0.1\text{--}10 \text{ Hz}$		0.8		μV_{pp}
Input bias current noise				
Noise density, $f = 100 \text{ Hz}$		2.5		$\text{fA}/\sqrt{\text{Hz}}$
Current noise, $\text{BW} = 0.1\text{--}10 \text{ Hz}$		48		fA_{pp}

Broadband current noise component

$$\text{BW}_n = f_H \cdot K_n \quad (2.1)$$

$$\text{BW}_n = (158 \text{ kHz}) \cdot (1.57) = 248 \text{ kHz}$$

$$I_{\text{nBB}} = i_{\text{nBB}} \sqrt{\text{BW}_n} \quad (2.7)$$

$$I_{\text{nBB}} = \left(2.5 \frac{\text{fA}}{\sqrt{\text{Hz}}} \right) \cdot \sqrt{248 \text{ kHz}} = 1.244 \text{ pA}_{\text{RMS}}$$

$$I_{\text{nf}} = 0 \text{ A} \quad \text{No } 1/f \text{ noise}$$

$$I_n = \sqrt{I_{\text{nf}}^2 + I_{\text{nBB}}^2} \quad (2.11)$$

$$I_n = 1.244 \text{ pA}_{\text{RMS}}$$

$$R_{\text{eq}} = R_1 \parallel R_f$$

$$R_{\text{eq}} = \frac{R_f \cdot R_1}{R_f + R_1} = \frac{(100 \text{ k}\Omega) \cdot (1 \text{ k}\Omega)}{100 \text{ k}\Omega + 1 \text{ k}\Omega}$$

$$E_{\text{n,i}} = I_n \cdot R_{\text{eq}} \quad (2.12)$$

$$E_{\text{n,i}} = (1.244 \text{ pA}_{\text{RMS}}) (0.99 \text{ k}\Omega) = 1.23 \text{ nV}_{\text{RMS}}$$

Figure 3.6: Conversion of current noise spectral density to equivalent input noise voltage

(see Figure 3.6) or in a table format (see Figure 3.7). The spectral density values and the closed loop bandwidth are used to compute the total input voltage noise. Example 3.1 shows how the total input noise is computed using the formulas introduced previously.

Thermal noise (resistor noise) component

$$BW_n = f_H \cdot K_n \quad (2.1)$$

$$BW_n = (158 \text{ kHz}) \cdot (1.57) = 248 \text{ kHz}$$

$$R_{eq} = R_1 \parallel R_f$$

$$R_{eq} = \frac{R_f \cdot R_1}{R_f + R_1} = \frac{(100 \text{ k}\Omega) \cdot (1 \text{ k}\Omega)}{100 \text{ k}\Omega + 1 \text{ k}\Omega} \quad (2.8)$$

$$E_{n_r} = \sqrt{4 \cdot k \cdot T \cdot R_{eq} \cdot \Delta f} \quad (2.13)$$

$$E_{n_r} = \sqrt{4 \left(1.38 \cdot 10^{-23} \text{ J/K} \right) (298 \text{ K}) \cdot (0.99 \text{ k}\Omega) \cdot (248 \text{ kHz})} = 2010 \text{ nV}_{\text{RMS}}$$

Since the total voltage noise is $E_{n_v} = 2497 \text{ nV}_{\text{RMS}}$, the resistor noise voltage ($2010 \text{ nV}_{\text{RMS}}$) is significant.

Figure 3.7: Conversion of resistor noise to equivalent input noise voltage

3.4 Compute Total Op-Amp Voltage Noise Contribution

3.4.1 Compute Total Op-Amp Current Noise Contribution

Now we need to convert the current noise to an equivalent referred-to-the-input (RTI) voltage noise. First, we will convert the current noise spectral density to a current source. This current source is multiplied by an equivalent input resistance to compute input voltage noise. It should be noted that the $1/f$ calculation is not required for this example because the amplifier is a JFET input. JFET amplifiers generally do not have $1/f$ current noise. This procedure is summarized in Figure 3.6.

3.5 Compute Total Thermal Noise Contribution

Figure 3.7 illustrates how RTI resistor noise is calculated. Note that for this example, the resistor noise is similar in magnitude to the op-amp noise and so it will significantly contribute to the output noise.

3.6 Combine All the Noise Sources and Compute Peak-to-Peak Output

Now that we have computed all the noise components, we can determine the total noise RTI. This result will be multiplied by the noise gain to compute the noise referred to the output.

Voltage noise form op-amp RTI:

$$E_{n_v} = 2497 \text{ nV}_{\text{RMS}} \quad \text{Figure 3.5}$$

Current noise form op-amp RTI:

$$E_{n_i} = 1.23 \text{ nV}_{\text{RMS}} \quad \text{Figure 3.6}$$

Resistor noise form op-amp RTI:

$$E_{n_r} = 2010 \text{ nV}_{\text{RMS}} \quad \text{Figure 3.7}$$

Total RMS noise RTI:

$$E_{n_{in}} = \sqrt{E_{n_v}^2 + E_{n_i}^2 + E_{n_r}^2} \quad (2.14)$$

$$E_{n_{in}} = \sqrt{(2497 \text{ nV}_{\text{RMS}})^2 + (1.23 \text{ nV}_{\text{RMS}})^2 + (2010 \text{ nV}_{\text{RMS}})^2} = 3205 \text{ nV}_{\text{RMS}}$$

Total RMS noise RTO:

$$E_{n_{out}} = E_{n_{in}} \cdot \text{Noise_Gain} \quad (2.15)$$

$$E_{n_{out}} = (3205 \text{ nV}_{\text{RMS}}) (101) = 324 \text{ } \mu\text{V}_{\text{RMS}}$$

Estimate total peak-to-peak noise RTO:

$$E_{n_{out_pp}} = E_{n_{out}} \cdot 6 \quad (2.16)$$

$$E_{n_{out_pp}} = (324 \text{ } \mu\text{V}_{\text{RMS}}) \cdot 6 = 1.94 \text{ mV}_{\text{pp}}$$

Figure 3.8: Computation of total peak-to-peak output noise

Finally, the conversion factor from Table 1.1 will be used to estimate the peak-to-peak output. [Figure 3.8](#) shows the details.

3.7 Example Calculation #2: Two-Stage Amplifier

Example noise calculation #2 will analyze the noise for a two-stage amplifier. The goal of this example is to show how the equations derived in Chapter 2 can be applied to a different circuit topology. This circuit has source impedance R_1 associated with V_{in} . The circuit also has two low-pass filters (R_4C_1 and R_7C_2) ([Figure 3.9](#)).

[Figure 3.10](#) shows the calculations for the total noise from the input amplifier's (U1) noise voltage source. This noise contains both $1/f$ and broadband noise. This noise does not include any effects of the second-stage amplifier (U2). The noise bandwidth in this example is limited by the two filters. Because the cut frequency (f_c) of the two filters is the same, they act as a single second-order filter ($K_n = 1.22$).

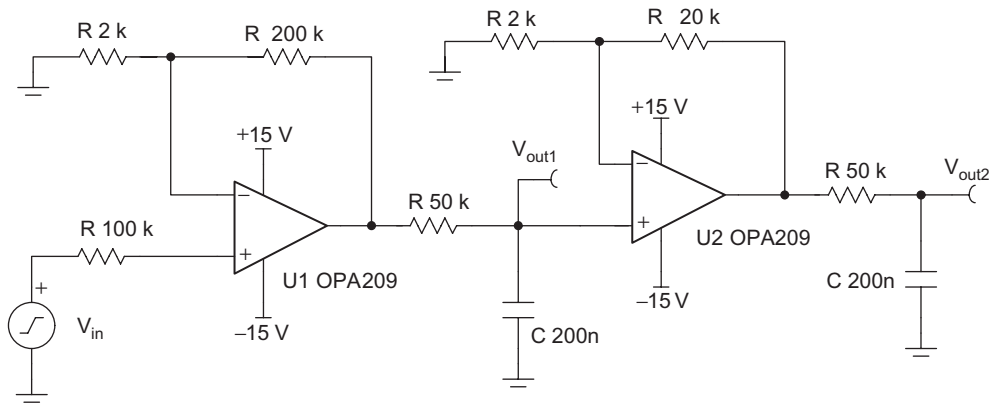


Figure 3.9: Two-stage amplifier for noise example

Gain_Bandwidth_Product := 18 MHz From data sheet

Noise_Gain := 101

Closed_Loop_Bandwidth := $\frac{\text{Gain_Bandwidth_Product}}{\text{Noise_Gain}} = 178.2 \times 10^3 \text{ Hz}$

$f_c := \frac{1}{2\pi \cdot (50 \text{ k}\Omega) \cdot (200 \text{ nF})} = 15.9 \text{ Hz}$

$f_c < \text{Closed_Loop_Bandwidth}$ so f_c is used for noise bandwidth

$\text{BW}_n := 1.22 f_c = 19.4 \text{ Hz}$

$e_n := 2.2 \frac{\text{nV}}{\sqrt{\text{Hz}}}$ From data sheet

$E_{\text{nBB}} := e_n \cdot \sqrt{\text{BW}_n} = 9.69 \times 10^{-9} \text{ V}$ Total broadband noise

$f_L := 0.1 \text{ Hz}$ $e_{\text{at}_f} := 20 \frac{\text{nV}}{\sqrt{\text{Hz}}}$ From data sheet

$e_{\text{fnorm}} := e_{\text{at}_f} \cdot \sqrt{f_L} = 6.32 \times 10^{-9} \text{ V}$

$E_{\text{nf}} := e_{\text{fnorm}} \cdot \sqrt{\ln \left(\frac{\text{BW}_n}{f_L} \right)} = 14.52 \times 10^{-9} \text{ V}$ Total 1/f noise

$E_{\text{n}_v} := \sqrt{E_{\text{nBB}}^2 + E_{\text{nf}}^2} = 17.46 \times 10^{-9} \text{ V}$ Total noise from voltage noise

Figure 3.10: Total noise from first-stage op-amp noise voltage

Figure 3.11 shows the calculations for the total noise from the input amplifier's (U1) noise current source. The current noise flows through both the feedback network and the source impedance R_1 . This noise does not include any effects of the second-stage amplifier (U2).

Figure 3.12 shows the calculations for the total noise from the input amplifier's (U1) resistor noise. Both the feedback network (R_2 and R_3) and the source impedance (R_1) contribute to the noise. This noise does not include any effects of the second-stage amplifier (U2).

Figure 3.13 shows the combination of the current voltage and resistor noise for the input stage (U1). In this example, the current noise is the dominant noise source. This noise does not include any effects of the second-stage amplifier (U2).

$$\begin{aligned}
 i_{BB} &:= 0.5 \frac{\text{pA}}{\sqrt{\text{Hz}}} && \text{From data sheet} \\
 I_{nBB} &:= i_{BB} \cdot \sqrt{BW_n} = 2.203 \times 10^{-12} \text{ A} \\
 R_1 &:= 2 \text{ k}\Omega && R_f := 200 \text{ k}\Omega \\
 R_{eq} &:= \frac{R_1 \cdot R_f}{R_1 + R_f} = 1.98 \times 10^3 \Omega \\
 E_{n_{i1}} &:= I_{nBB} \cdot R_{eq} = 4.363 \times 10^{-9} \text{ V} && \text{Current noise flowing through feedback network} \\
 R_{in} &:= 100 \text{ k}\Omega \\
 E_{n_{i2}} &:= I_{nBB} \cdot R_{in} = 220.323 \times 10^{-9} \text{ V} && \text{Current noise flowing through input resistance} \\
 E_{n_i} &:= \sqrt{E_{n_{i1}}^2 + E_{n_{i2}}^2} = 220.366 \times 10^{-9} \text{ V} && \text{Total noise from current noise}
 \end{aligned}$$

Figure 3.11: Total noise from first-stage noise current

$$\begin{aligned}
 k_b &:= 1.38 \cdot 10^{-23} \text{ J/K} \\
 T_a &:= 298 \text{ K} \\
 E_{n_{r1}} &:= \sqrt{4 \cdot k_b \cdot T_a \cdot R_{eq} \cdot BW_n} = 25.1 \times 10^{-9} \text{ V} \\
 E_{n_{r2}} &:= \sqrt{4 \cdot k_b \cdot T_a \cdot R_{in} \cdot BW_n} = 178.7 \times 10^{-9} \text{ V} \\
 E_{n_r} &:= \sqrt{E_{n_{r1}}^2 + E_{n_{r2}}^2} = 180.5 \times 10^{-9} \text{ V} && \text{Total noise from resistor noise}
 \end{aligned}$$

Figure 3.12: Total noise from first-stage resistor noise

The second-stage calculations are the same as the first-stage calculations except that the feedback network and source resistance have different values. Figure 3.14 summarizes the results. Figure 3.15 shows the calculation for the total noise from both stages. Note that the second-stage noise does not have a significant effect on the total noise. This is because the input-stage noise was amplified by a factor of 101 before combining with the second-stage noise. In general, the input-stage noise will dominate. This is especially true when the first stage is in high gain. Because the input stage will dominate, engineers will often use more expensive low-noise amplifiers for the input stage and less expensive commodity amplifiers for the output stage.

$$\begin{aligned}
 E_{n_i} &= 220.4 \times 10^{-9} \text{ V} \\
 E_{n_v} &= 17.5 \times 10^{-9} \text{ V} \\
 E_{n_r} &= 180.5 \times 10^{-9} \text{ V} \\
 E_{n_{in}} &:= \sqrt{E_{n_i}^2 + E_{n_v}^2 + E_{n_r}^2} = 285.4 \times 10^{-9} \text{ V} \quad \text{Total noise in the input stage (U1).}
 \end{aligned}$$

Figure 3.13: Total noise in the first stage

$$\begin{aligned}
 E_{n_i} &= 110.234 \times 10^{-9} \text{ V} \\
 E_{n_v} &= 17.456 \times 10^{-9} \text{ V} \\
 E_{n_r} &= 128.65 \times 10^{-9} \text{ V} \\
 E_{n_{in}} &:= \sqrt{E_{n_i}^2 + E_{n_v}^2 + E_{n_r}^2} = 170.3 \times 10^{-9} \text{ V} \quad \text{Total second stage noise from current, voltage, and resistance noise}
 \end{aligned}$$

Figure 3.14: Total noise from second-stage op-amp noise voltage

$$\begin{aligned}
 E_{n_{in1}} &:= 285.4 \text{ nV} \\
 \text{Gain1} &:= 101 \quad \text{Gain2} := 11 \\
 E_{out1} &:= E_{n_{in1}} \cdot \text{Gain1} \cdot \text{Gain2} = 317.079 \times 10^{-6} \text{ V} \quad \text{Total output from input stage noise} \\
 E_{n_{in2}} &:= 170.3 \text{ nV} \\
 E_{out2} &:= E_{n_{in2}} \cdot \text{Gain2} = 1.873 \times 10^{-6} \text{ V} \quad \text{Total output from second stage noise} \\
 E_{n_{out_total}} &:= \sqrt{E_{out1}^2 + E_{out2}^2} = 317.085 \times 10^{-6} \text{ V} \quad \text{Total output noise for example \#2 including all noise sources.}
 \end{aligned}$$

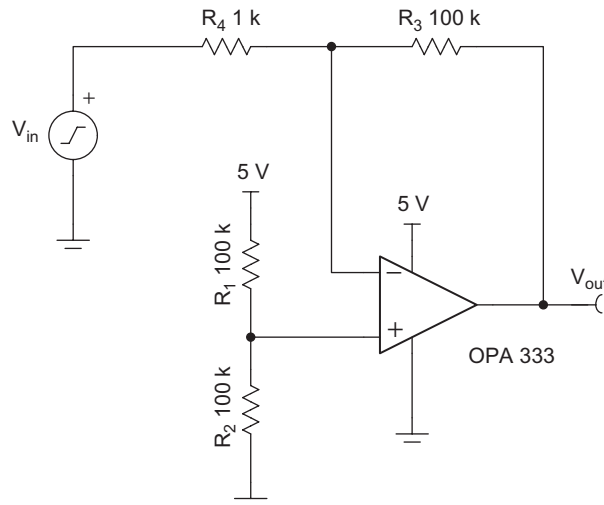
Figure 3.15: Total noise from second-stage op-amp noise voltage

Chapter Summary

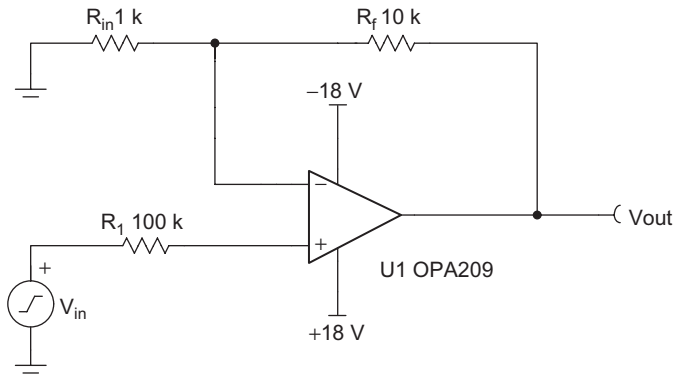
- The gain bandwidth of an amplifier will set the noise bandwidth if no other filters exist.
- A simple RC filter can be used to set the noise bandwidth.
- In two-stage amplifiers, the input stage will dominate.
- Hand calculations give insight where a source dominates (voltage noise, current noise, or resistor noise).
- Current noise or resistor noise may dominate for amplifiers with large source resistances.
- Voltage noise typically dominates for amplifiers with low source resistance and low resistance feedback networks.

Questions

- 3.1
- What is the noise bandwidth for the circuit below?
 - What is the total output noise from op-amp voltage noise?
 - What is the total output noise from op-amp current noise?
 - What is the total output noise from op-amp resistor noise?
 - What is the total combined output noise from all noise sources?



- 3.2
- What is the noise bandwidth for the circuit below?
 - What is the total output noise from op-amp voltage noise?
 - What is the total output noise from op-amp current noise?
 - What is the total output noise from op-amp resistor noise?
 - What is the total combined output noise from all noise sources?



Further Reading

Hogg R.V., Tanis E.A., 1988, Probability and Statistical Inference, third ed. Macmillan, London.

Motchenbacher C.D., Connelly J.A., 1993, Low-Noise Electronic System Design, Wiley-Interscience, New York.

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Answers to Questions

Chapter 1

$$1.1 \quad E_n = \sqrt{4kTR\Delta f} = \sqrt{4(1.381 \times 10^{-23} \text{ J/K})(298.15 \text{ K})(10 \text{ k}\Omega)(10 \text{ kHz})} \\ = 1.283\text{-}\mu\text{V RMS}$$

$$1.2 \quad E_{n_pp} = 6\sigma = 6(1.283\mu\text{V}) = 7.698\mu\text{V}_{pp}$$

$$1.3 \quad E_n = \sqrt{4kTR} = \sqrt{4(1.381 \times 10^{-23} \text{ J/K})(298.15 \text{ K})(10 \text{ k}\Omega)} = 128.3(\text{nV}/\sqrt{\text{Hz}})$$

$$1.4 \quad E_{nT} = \sqrt{E_{n1}^2 + E_{n2}^2} = \sqrt{(10\text{mV})^2 + (5\text{mV})^2} = 11.18\text{-mV RMS}$$

$$1.5 \quad E_{nT} = \sqrt{E_{n1}^2 + E_{n2}^2 + 2CE_{n1}E_{n2}} = \sqrt{(10\text{mV})^2 + (5\text{mV})^2 + 2(-1)(10\text{mV})(5\text{mV})} \\ = 5\text{-mV RMS}$$

$$1.6 \quad E_{\text{RMS}} = \frac{E_{npp}}{6} = \frac{25\text{mV}_{pp}}{6} = 4.167\text{-mV RMS}$$

Chapter 2

$$2.1 \quad BW_n = f_H K_n = (1\text{kHz})(1.57) = 1.57\text{kHz}$$

2.2

$$a. \quad e_{f\text{norm}} = e_{at_f} \sqrt{f} = (60\text{nV}/\sqrt{\text{Hz}}) \sqrt{0.1\text{Hz}} = 18.97\text{nV}$$

$$E_{nf} = e_{f\text{norm}} \sqrt{\ln\left(\frac{f_H}{f_L}\right)} = (18.97\text{nV}) \sqrt{\ln\left(\frac{1\text{kHz}}{0.1\text{Hz}}\right)} = 57.6\text{-nV RMS}$$

$$b. \quad BW_n = f_H K_n = (1\text{kHz})(1.57) = 1.57\text{kHz}$$

$$E_{nBB} = e_{BB} \sqrt{BW_n} = (3.8\text{nV}/\sqrt{\text{Hz}}) \sqrt{1.57\text{kHz}} = 150.6\text{-nV RMS}$$

2.3

$$a. \quad R_{eq} = \frac{R_f R_l}{R_f + R_l} = \frac{(1M\Omega)(100k\Omega)}{1M\Omega + 100k\Omega} = 90.8k\Omega$$

$$E_{n_r} = \sqrt{4kTR\Delta f} = \sqrt{4(1.381 \times 10^{-23} \text{ J/K})(298.15 \text{ K})(90.8k\Omega)(5 \text{ kHz})} \\ = 2.734\text{-}\mu\text{V RMS}$$

$$b. \quad I_{nBB} = i_{nBB} \sqrt{BW_n} = (5 \text{ fA}/\sqrt{\text{Hz}}) \sqrt{(5 \text{ kHz})} = 354\text{-fA RMS}$$

$$I_n = I_{nBB} \quad \text{Because no flicker component is specified}$$

$$E_{n_i} = I_n R_{eq} = (354 \text{ fA})(90.8k\Omega) = 32.1\text{-nV RMS}$$

$$2.4 \quad E_{n_{in}} = \sqrt{E_{n_i}^2 + E_{n_v}^2 + E_{n_r}^2} = \sqrt{(100\mu\text{V})^2 + (150\mu\text{V})^2 + (75\mu\text{V})^2} \\ = 195\text{-}\mu\text{V RMS}$$

Chapter 3

3.1

$$a. \quad \text{Closed_Loop_Bandwidth} = \frac{\text{Gain_Bandwidth_Product}}{\text{Noise_Gain}} = \frac{350 \text{ kHz}}{101} = 3.465 \text{ kHz}$$

$$BW_n = f_H K_n = (3.465 \text{ kHz})(1.57) = 5.44 \text{ kHz}$$

$$b. \quad E_{nBB} = e_{BB} \sqrt{BW_n} = 55 \text{ nV}/\sqrt{\text{Hz}} \sqrt{(5.44 \text{ kHz})} = 4.057\text{-}\mu\text{V RMS}$$

$$c. \quad I_{nBB} = i_{nBB} \sqrt{BW_n} = (100 \text{ fA}/\sqrt{\text{Hz}}) \sqrt{(5.44 \text{ kHz})} = 7.38\text{-pA RMS}$$

$$R_{eq1} = \frac{R_f R_l}{R_f + R_l} = \frac{(100k\Omega)(1k\Omega)}{100k\Omega + 1k\Omega} = 0.99k\Omega$$

$$E_{n_{i1}} = I_n R_{eq} = (7.38 \text{ pA})(0.99k\Omega) = 7.31\text{-nV RMS}$$

$$R_{eq2} = \frac{R_f R_l}{R_f + R_l} = \frac{(100k\Omega)(100k\Omega)}{100k\Omega + 100k\Omega} = 50k\Omega$$

$$E_{n_{i2}} = I_n R_{eq} = (7.38 \text{ pA})(50k\Omega) = 369\text{-nV RMS}$$

$$E_{n_i} = \sqrt{E_{n_{i1}}^2 + E_{n_{i2}}^2} = \sqrt{(7.31 \text{ nV})^2 + (369 \text{ nV})^2} = 369.1 \text{ nV}$$

$$\begin{aligned} \text{d. } E_{n_{r1}} &= \sqrt{4kTR\Delta f} = \sqrt{4(1.381 \times 10^{-23} \text{ J/K})(298.15 \text{ K})(0.99 \text{ k}\Omega)(5.44 \text{ kHz})} \\ &= 298\text{-nV RMS} \end{aligned}$$

$$\begin{aligned} E_{n_{r2}} &= \sqrt{4kTR\Delta f} = \sqrt{4(1.381 \times 10^{-23} \text{ J/K})(298.15 \text{ K})(50 \text{ k}\Omega)(5.44 \text{ kHz})} \\ &= 2.117\text{-}\mu\text{V RMS} \end{aligned}$$

$$E_{n_i} = \sqrt{E_{n_{r1}}^2 + E_{n_{r2}}^2} = \sqrt{(298 \text{ nV})^2 + (2.117 \mu\text{V})^2} = 2.138\text{-}\mu\text{V RMS}$$

$$\begin{aligned} \text{e. } E_{n_{in}} &= \sqrt{E_{n_i}^2 + E_{n_v}^2 + E_{n_r}^2} = \sqrt{(369.1 \text{ nV})^2 + (4.057 \mu\text{V})^2 + (2.138 \mu\text{V})^2} \\ &= 4.601\text{-}\mu\text{V RMS} \end{aligned}$$

$$E_{n_{out}} = \text{Noise_Gain} \cdot E_{n_{in}} = 101(4.601 \mu\text{V}) = 464\text{-}\mu\text{V RMSRTO}$$

3.2

$$\begin{aligned} \text{a. } \text{Closed_Loop_Bandwidth} &= \frac{\text{Gain_Bandwidth_Product}}{\text{Noise_Gain}} = \frac{18 \text{ MHz}}{11} \\ &= 1.636 \text{ MHz} \end{aligned}$$

$$\text{BW}_n = f_H K_n = (1.636 \text{ MHz})(1.57) = 2.569 \text{ MHz}$$

$$\text{b. } E_{n_{BB}} = e_{BB} \sqrt{\text{BW}_n} = 2.2 \text{ nV}/\sqrt{\text{Hz}} \sqrt{(2.569 \text{ MHz})} = 3.526\text{-}\mu\text{V RMS}$$

$$e_{f_{\text{norm}}} = e_{at_f} \sqrt{f} = 20 \text{ nV}/\sqrt{\text{Hz}} \sqrt{0.1 \text{ Hz}} = 6.325 \text{ nV}$$

$$E_{nf} = e_{f_{\text{norm}}} \sqrt{\ln\left(\frac{f_H}{f_L}\right)} = (6.325 \text{ nV}) \sqrt{\ln\left(\frac{2.569 \text{ MHz}}{0.1 \text{ Hz}}\right)} = 24.3\text{-nV RMS}$$

$$E_{n_v} = \sqrt{E_{nf}^2 + E_{n_{BB}}^2} = \sqrt{(24.3 \text{ nV})^2 + (3.526 \mu\text{V})^2} = 3.526\text{-}\mu\text{V RMS}$$

$$\text{c. } I_{n_{BB}} = i_{n_{BB}} \sqrt{\text{BW}_n} = (500 \text{ fA}/\sqrt{\text{Hz}}) \sqrt{(2.569 \text{ MHz})} = 801.4\text{-pA RMS}$$

$$I_{f_{\text{norm}}} = I_{at_f} \sqrt{f} = 3.75 \text{ pA}/\sqrt{\text{Hz}} \sqrt{0.1 \text{ Hz}} = 1.186 \text{ pA}$$

$$I_{nf} = I_{f_{\text{norm}}} \sqrt{\ln\left(\frac{f_H}{f_L}\right)} = (1.186 \text{ pA}) \sqrt{\ln\left(\frac{2.569 \text{ MHz}}{0.1 \text{ Hz}}\right)} = 4.9\text{-pA RMS}$$

$$I_n = \sqrt{I_{nf}^2 + I_{n_{BB}}^2} = \sqrt{(4.9 \text{ pA})^2 + (801.4 \text{ pA})^2} = 801.4\text{-pA RMS}$$

$$R_{eq1} = \frac{R_f R_i}{R_f + R_i} = \frac{(10 \text{ k}\Omega)(1 \text{ k}\Omega)}{10 \text{ k}\Omega + 1 \text{ k}\Omega} = 0.909 \text{ k}\Omega$$

$$E_{n_{i1}} = I_n R_{eq1} = (801.4 \text{ pA})(0.99 \text{ k}\Omega) = 7.31\text{-nV RMS}$$

$$E_{n_{i2}} = I_n R_{in} = (801.4 \text{ pA})(100 \text{ k}\Omega) = 80.1\text{-}\mu\text{V RMS}$$

$$E_{n_i} = \sqrt{E_{n_{i1}}^2 + E_{n_{i2}}^2} = \sqrt{(7.31 \text{ nV})^2 + (80.1 \mu\text{V})^2} = 80.1\text{-}\mu\text{V RMS}$$

d. $E_{n_{r1}} = \sqrt{4kTR\Delta f}$
 $= \sqrt{4(1.381 \times 10^{-23} \text{ J/K})(298.15 \text{ K})(0.909 \text{ k}\Omega)(5.44 \text{ kHz})} = 285\text{-nV RMS}$
 $E_{n_{r2}} = \sqrt{4kTR\Delta f} = \sqrt{4(1.381 \times 10^{-23} \text{ J/K})(298.15 \text{ K})(100 \text{ k}\Omega)(5.44 \text{ kHz})}$
 $= 2.99\text{-}\mu\text{V RMS}$
 $E_{n_r} = \sqrt{E_{n_{r1}}^2 + E_{n_{r2}}^2} = \sqrt{(285\text{ nV})^2 + (2.99\text{ }\mu\text{V})^2} = 3\text{-}\mu\text{V RMS}$

e. $E_{n_{in}} = \sqrt{E_{n_i}^2 + E_{n_v}^2 + E_{n_r}^2} = \sqrt{(80.1\text{ }\mu\text{V})^2 + (3.526\text{ }\mu\text{V})^2 + (3\text{ }\mu\text{V})^2}$
 $= 80.2\text{-}\mu\text{V RMS}$
 $E_{n_{out}} = \text{Noise_Gain} \cdot E_{n_{in}} = 11(80.2\text{ }\mu\text{V}) = 882\text{-}\mu\text{V RMSRTO}$

Chapter 4

4.1

